

Senior Data Scientist & Software Developer at Urban Analytics startup out of **Harvard Innovation Labs**. Specialising in digital twins, advanced ML, econometrics and complex systems analysis | *MSc Geospatial Data Science, MA Economics*

TECHNICAL SKILLS

Language & Libraries | *R, Javascript, SQL, Python* (XGBoost, PyTorch, TensorFlow, sklearn, SciPy, NetworkX, PySAL)
Software | AWS, GCP, ClaudeCode, Supabase, Google Earth Engine, GitHub, QGIS, CARTO, Snowflake, PowerBI, Gephi
Skills | Software Development, Digital Twins, Climate Modelling, Monte Carlo Simulation, Gradient Descent, Causal Inference, Neural Networks, Remote Sensing, Urban Simulation, Data Architecture, ML, Project Management

EDUCATION

M.Sc Spatial Data Science | **University College London** | *Distinction* **2023 – 2024**

- Specialised in applying complexity science and ML to geospatial problems at *The Bartlett School of Built Environment*
- Awarded funding (£500) from *Space Syntax Ltd.* and the *Consumer Data Research Centre* to write MSc dissertation

M.A (Hons) Economics | **University of Edinburgh** | *First Class* **2016 – 2020**

- **Dissertation:** *The impact of urban sprawl and modern zoning laws on indigenous communities in Western Canada*

International Baccalaureate | **United World College Maastricht** | *Summa Cum Laude 40/45* **2013 – 2015**

- Awarded 2 year Academic Scholarship: £40 000 | 96th percentile internationally

WORK EXPERIENCE

Aretian Urban Analytics | **Senior Data Scientist & Software Developer** | *London, UK* **Nov 2025 – Present**

- Applying Economic Complexity methods (PCI and PDI) to advise on how cities can develop more resilient economies
- Measured infrastructure & street network structure to assess how physical layout relates to economic performance
- Using causal inference, systems analytics and geospatial ML to inform economic & infrastructure strategy for cities
- Developed multi-hazard climate modelling platform for Barcelona, integrating 130 spatial indicators across 216,000+ hexagons to produce Heat & Flood Vulnerability Indices helping inform climate & infrastructure decisions
- Building data pipelines in AWS for city-scale digital twins, combining 100+ variables into urban performance model
- Large scale web-scraping: Developed automated pipelines to extract and clean financial and infrastructure data from unstructured PDF reports, linking to regional economic indicators for cross-city comparison
- Automating routine tasks, data pipelines and data engineering processes using LLMs, Agents and Claude Code
- Spatial and network visualisation for a variety of urban analytics and econometric datasets for client deliverables
- Co-authored research paper which models the innovation ecosystem of the Boston Cambridge Somerville Region

University of Oxford | **Inst. for New Economic Thinking** | **Volunteer Researcher** | *Oxford, UK* **Apr 2026 – Sep 2026**

- Contributing to Oxford Martin School's Programme on *Forecasting Technological Change* w/ Prof. Francis Lafond & Dr. Amir Akther. To be credited in public database on clean tech learning curves
- Researching how clean technologies improve & become cheaper over time helping inform UK climate policy
- Developed & validated learning curve models to forecast future cost reductions of climate critical UK technologies

Norman Foster Institute | **City Science Research Assistant** | *Madrid, Spain* **Jan 2025 – Dec 2025**

- Awarded **€52 000** grant to research Spatial Data Science & Sustainable Development in Cape Town, South Africa
- Awarded *Masters in Sustainable Cities* from the Autonomous University of Madrid & the Norman Foster Institute
- Developing *Densification Potential Model* to identify areas of unrealised economic potential in Cape Town. Tool built

- in Google Earth Engine using LIDAR, housing capacity, population density, Land Use Zoning in Python & JavaScript
- Developed Cape Town *Public Space Qualification Tool* to assess their quality in line with municipal development plan

Space Syntax Ltd. | Data Science Researcher | London, UK

Apr 2024 – Sept 2024

- **Dissertation Title:** *Using unsupervised machine learning methods to identify multiscale centrality indicators*
- Authored research paper in conjunction with Space Syntax Ltd. which predicts the location of economic activity based on the mathematical layout of UK street networks and how this model changes at different scales
- Examined the layout of transportation & street networks to predict optimal business locations
- Offered full time Spatial Data Scientist role at Space Syntax Ltd. based on skillset and quality of work

Baringa Partners | Management Consultant (Data & Analytics) | London, UK

Mar 2021 – Aug 2023

Designed organisational structure of Vodafone consumer division for future merger. Delivered ~ £10m OpEx reduction

- Spearheaded design of Data & Analytics team. Collaborated closely w/ senior leadership through iterative workshops
- Developed 6-month implementation & communication plan to ensure seamless transition to new organisation structure

Developed technology & AI strategy for Magic Circle law firm *Freshfields Bruckhaus Deringer*

- Designed 2 data pipelines, which ingest legal data, predict likelihood of winning dispute, legal costs & pay-out amounts

Designed data architecture which helped *Thomson Reuters* develop a more targeted marketing and sales strategy

- Used data visualisation software (PowerBI) to represent client's sales performance and profitability over time

Agile Project Manager: Developed data pipeline for largest music licensing company in the UK

- Built data pipeline to increase data ingestion speed by 500% and produce insights from unutilised metadata

Strategist working on the largest supply chain automation in UK Publishing history (*Penguin Random House*)

- Designed supply chain of UK's largest book publisher to accommodate COVID demand spike & a recent acquisition
- Developed & led 10+ training sessions to ensure new analytics software was correctly utilised: staff now had fewer touchpoints, more storage recommendations by SKU & new conveyer belts

O5 Partners | Management Consultant (Intern) | London, UK

Sept 2020 – Mar 2021

- Worked on a restructuring and organisational design project for *National Grid* (£14b revenue utilities company)

Boston Consulting Group | Mobility Analyst | London, UK

Sept 2018 – Oct 2019

- Student researcher at BCG where I focussed on Autonomous Vehicles, Smart Cities & Intermodal Applications

ACHIEVEMENTS

- €52 000 Research grant from the Norman Foster Institute to study Sustainable Cities and Spatial Data Science | 01/2025
- 3rd Place | Dissertation awarded 3rd at the Society for Location Analysis *UK Masters Dissertation Awards* | 11/2024
- 1st Place | Centre for Advanced Spatial Analysis Hackathon w/ West Midlands Fire Service | £2000 Cash Prize | 06/2024
- 1st Place | Boston Consulting Group *London University Case Study Competition* | £500 Cash Prize | 09/2019

PUBLICATIONS

Norman Foster Institute (2026), *Research on Cities: Volume II*. Ivory Press & Norman Foster Foundation. Available here:

<https://ivorypress.com/en/product/research-on-cities-volume-ii/>.

[GitHub Profile](#)

[Personal Website](#)